MIL-F-10589A
10 JUNE 1952
SUPERSEDING
MIL-F-10589(ORD)
9 OCTOBER 1950

MILITARY SPECIFICATION

FUZE, ROCKET, BASE DETONATING, M404A1, LOADING, ASSEMBLING AND PACKING

This specification was approved by the Departments of the Army, the Navy, and the Air Force for the use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers one type of fuze which is intended for use in the 3.5-inch HEAT rocket, M28A2.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following drawings and all drawings and specifications referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

DRAWINGS

ORDNANCE CORPS

73–10–35—Fuze, Rocket, BD, M404A1, Assembly.
81– 3–30—Machine, Jolt Testing.
81– 3–35—Machine, Jumble Testing
81– 3–37—Fixture, Jumble Testing.

(Copies of specifications and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.**—Material and parts shall be in accordance with applicable drawings and specifications.

3.2 The fuzes shall meet all requirements specified on the drawings and in applicable specifications.

3.3 **Jolt and jumble.**—The fuzes shall withstand 1,750 jolts in each of the three positions of the jolt machine and, after completion of the jolt test, the fuzes used in the jolt test shall withstand 3,600 complete revolutions in the jumble machine, without having any:

explosive elements explode,
setback sleeve displaced to its armed position
ejection pin release upon removal of the safety band assembly,
parts broken or separated from the assembly or any sub assembly,
parts deformed or displaced in such a manner as to make the fuze

unsafe to handle or dangerous to fire.

A lot of fuzes will be considered to meet these requirements if samples selected and tested in accordance with 4.5 meet the acceptance provisions prescribed therein.

3.4 **Proving ground tests.**

3.4.1 No premature burst shall occur (see 6.2).

3.4.2 The fuzes shall cause the rocket to burst with high order detonation when fired for impact against a wood-bursting screen.

3.4.3 The fuzes shall cause the rocket to burst with high order detonation when fired for graze impact.

3.4.4 A lot of fuzes will be considered to meet these requirements if samples selected

983349—52—(629)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:15Z
Check the source to verify that this is the current version before use.

and tested in accordance with 4.6 meet the acceptance provisions prescribed therein.

3.5 Workmanship.—All parts shall be free of chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot. — A lot shall consist of twenty thousand fuzes, maximum. A smaller quantity may constitute a lot at the discretion of the inspector. Two thousand plus or minus 500 shall constitute the first lot manufactured on any contract, except when a new contract immediately follows the manufacture of satisfactory material of the same character. Each lot shall contain metal parts from not more than one lot.

4.2 Sampling.—All samples shall be selected by the inspector. Twenty-five fuzes shall be selected from each lot for tests as specified in 4.5 and 4.6. Sampling of component materials is prescribed in paragraphs covering tests of the material.

4.3 Inspection.—Inspection shall be as specified in Specification MIL–G–2550.

4.4 Determination of binder-lubricant content of pellets.—Ten booster pellets selected by the inspector from each lot before assembling the fuzes shall be tested as follows:

4.4.1 Remove the outer surface from the sample booster pellets to obtain a composite sample of about 5 grams (gms.) from the interior of the booster pellets. Transfer this weighed portion of approximately 5 gms. of the sample to a tared filtering crucible and extract with four 10 milliliters (ml.) portions of acetone, maintaining each portion in contact with the residue for approximately 5 minutes before applying suction, breaking up any lumps which may persist, and washing down the sides of the crucible with each 10 ml. portion of acetone to insure complete removal of the tetryl. Aspirate the residue until the odor of acetone is no longer detectable and dry at 100° to 105°C. for 1 hour. Calculate the weight of residue to percentage of binder-lubricant in the sample.

4.4.2 *Retest.*—A retest, using double the number of samples used in the original test, may be made if requested by the contractor, provided the analysis indicates that the binder-lubricant content exceeds the percentage specified on the drawing by not more than 0.25.

4.5 Jolt and jumble tests of fuzes.

4.5.1 *Procedure.*

4.5.1.1 *Jolt.*—Assemble each of ten of the sample fuzes to an adapter (fig. 1) and then assemble to jolt testing machine, drawing 81–3–30. Give each fuze 1750 jolts in each of the three positions.

4.5.1.2 *Jumble.*—After completion of the jolt test, subject each of the fuzes used in the jolt test to a jumble test. Assemble in jumble fixture (fig. 2) and place, one at a time, in jumble machine, drawing 81–3–35. Give the machine 3,600 complete revolutions at the speed specified on drawing 81–3–35. Examine the fuzes upon completion of this test.

4.5.2 *Acceptance.*—The lot shall be considered acceptable provided that the 10 fuzes tested meet the requirements of 3.3. The lot shall be considered eligible for retest under the provisions of 4.5.3 provided none of the explosive elements explode, no setback sleeve is displaced to its armed position, and no ejection pin is released upon removal of the safety band assembly, and if:

one fuze has any parts broken or separated from the assembly or any subassembly but none in such a manner as to make the fuze unsafe to handle or dangerous to fire. The sealing disc in a maximum of 50 percent of

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:15Z
Check the source to verify that this is the current version before use.

the fuzes may be separate from the fuze without causing rejection of the lot, or

one fuze has any parts deformed or displaced but none in such a manner as to make the fuze unsafe to handle or dangerous to fire. If the limits for eligibility for retest are exceeded, the lot shall be rejected.

4.5.3 *Retest.*—A retest as provided for in 4.5.2 may be made if requested by the contractor, using 20 fuzes for the retest. The lot shall be considered acceptable on retest provided that the 20 fuzes tested meet the requirements of 3.3.

4.6 Proving ground tests.

4.6.1 *Procedure.*—Assemble fifteen of the sample fuzes in M28A2, 3.5-Inch HEAT Rockets, condition for at least 18 hours at 70°F., and fire from a launcher of latest design for which the rocket is standard, as follows:

4.6.1.1 Fire ten of the rockets against a 2-inch (1⅝ inch minimum) thick wood bursting screen. The screen shall be placed not less than 100 feet from the muzzle of the launcher and normal to the line of fire.

4.6.1.2 Fire five of the rockets with a launcher elevation of 6° for graze impact on firm ground. Record the range for each rocket.

4.6.2 *Acceptance.*—The lot shall be considered acceptable provided that in the phase being tested, i.e., screen impact or graze impact phase, the rockets fired and met the applicable requirements of 3.4. The lot shall be considered eligible for retest under the provisions of 4.6.3 provided no premature burst occurred, and if:

one fuze failed to cause the rocket to burst with high order detonation when fired for impact against a wood bursting screen, or

one fuze failed to cause the rocket to burst with high order detonation when fired for graze impact.

If the limits for eligibility for retest are exceeded, the lot shall be rejected.

4.6.3 *Retest.*—A retest as provided for in 4.6.2 may be made in the phase in which failures occur, if requested by the contractor using double the number of samples used in the original test. The lot shall be considered acceptable on retest provided that in the phase being tested, the rockets fired and met the applicable requirements of 3.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging. — Packaging shall be as specified on the applicable drawing.

AMMUNITION DATA CARD

Kind: Fuze, Rocket, Base Detonating M404A1 Ammunition Lot No._____

AIC Symbol_____ Quantity in Lot_____

Drg. No.*_____ Spec. No.*_____

Metal Parts Manufactured by_____ Lot No._____

Loaded and Assembled by_____ Date_____

Contract No._____ Order No._____

Packed_____in a box.

Remarks:_____

Certified by_____, Inspector.
Print here name of inspector, district, arsenal, or loading plant, and sub-district, where applicable.

* Include revision and amendments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:15Z
Check the source to verify that this is the current version before use.

5.2 Packing.—Packing shall be as specified on the applicable drawing.

5.3 Labeling and marking.—Labeling and marking for shipment shall be as specified on the applicable drawing.

5.4 Data cards.—Data cards, 5 by 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of accepted fuzes to provide the information specified below. Quantities (approximately 50 cards per lot) and distribution of data cards shall be in accordance with instructions issued by the Office, Chief of Ordnance.

6. NOTES

6.1 The contract should list all material to be furnished by the contractor.

6.2 Premature burst.—A premature burst is defined as a functioning of the bursting charge of the rocket head in the bore of the launcher or in flight before impact.

Notice.—This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, nor be used for any other purpose except when specifically authorized by the Chief of Ordnance.

Custodian:
Army—Ordnance Corps.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:15Z
Check the source to verify that this is the current version before use.

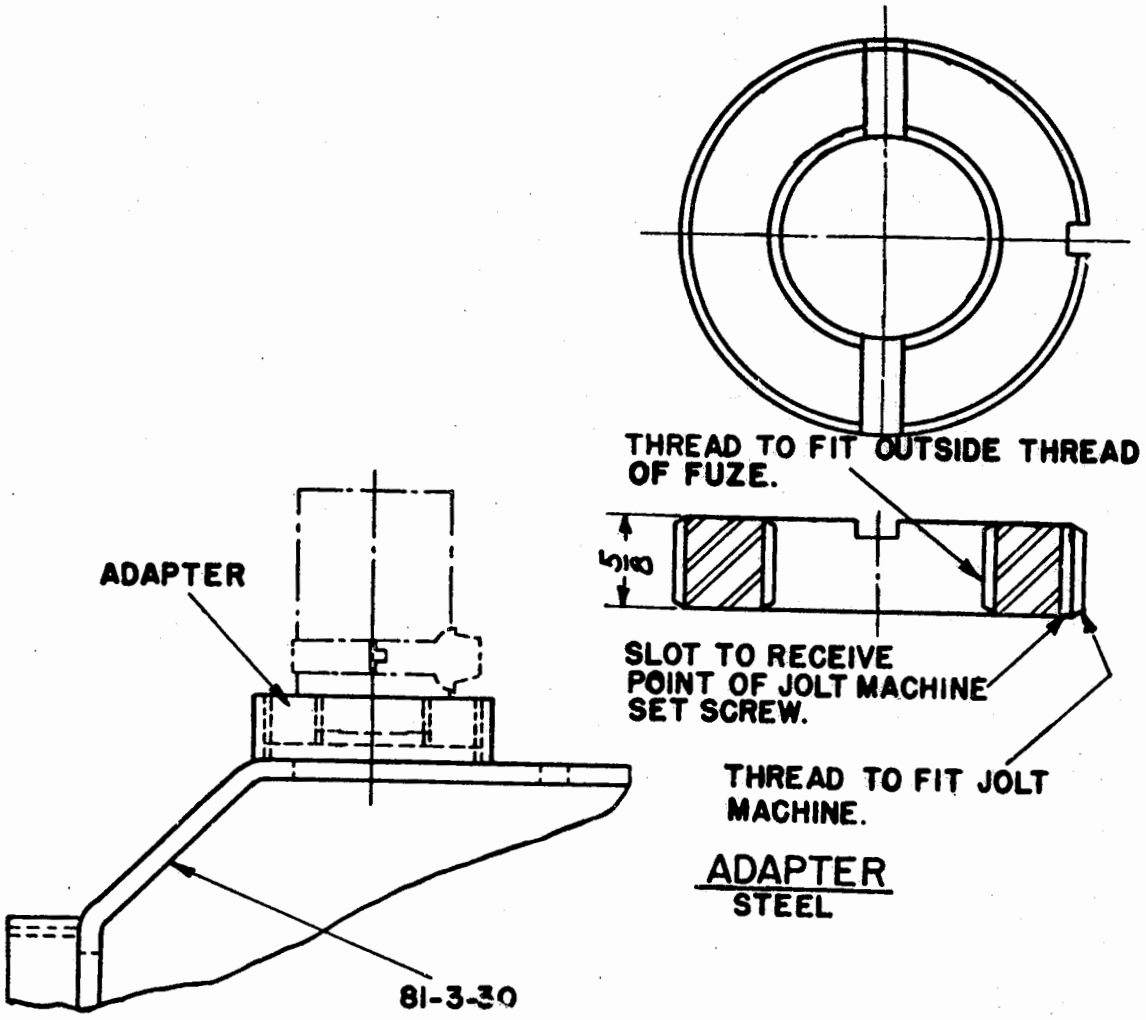


FIGURE 1.—*Part view of jolt machine.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:15Z
Check the source to verify that this is the current version before use.

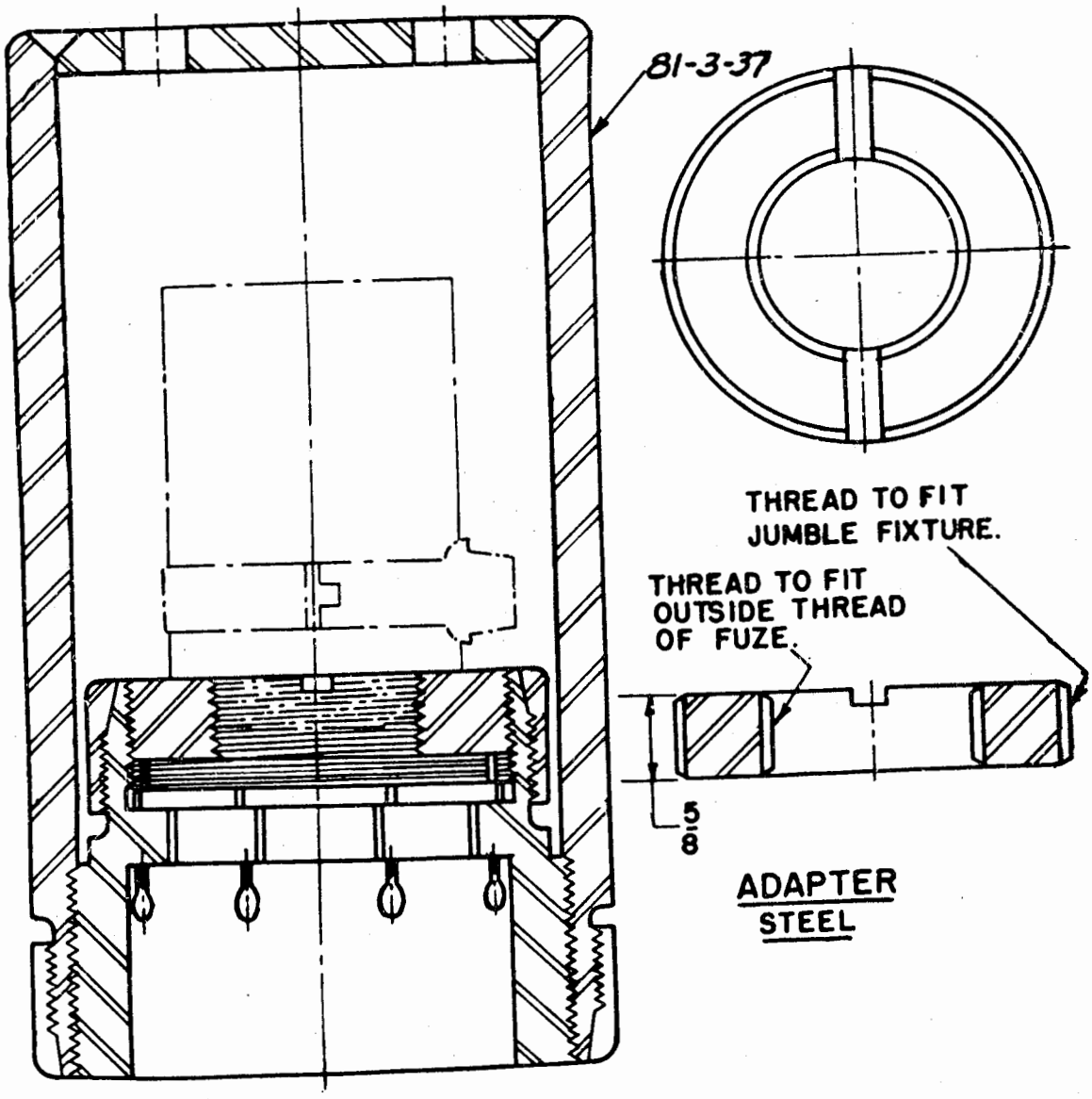


FIGURE 2.—*Jumble fixture.*

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T14:15Z
Check the source to verify that this is the current version before use.